

A3-PS015-TC241

Explainable Home and Last-Mile Connectivity Diagnosis

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Problem Description

•Current internet diagnostic tools measure speed, latency, packet loss, and DNS response.
•Users often struggle to understand why their internet connection performs poorly.
•PS-015 proposes a guided desktop tool for diagnosing connectivity issues.
•The system uses defined network measurements and transparent decision rules to identify likely problems.
•It generates clear diagnostic results and reports “INCONCLUSIVE” when evidence is insufficient.
•Identify local connectivity problems.
•Identify DNS resolution problems.
•Detect packet-loss conditions.
•Detect congestion or queueing conditions.
•Provide measurable evidence and an escalation report.

Background

•A connection can appear to be “slow” or “not working” for different reasons. A speed number alone does not explain whether the problem is the local gateway, DNS, packet loss, or congestion. PS-015 therefore uses a sequence of measurements rather than relying on one test.
•DNS in simple words: DNS is the system that converts a website name such as youtube.com into the IP address needed to reach that website. If DNS fails, the Internet connection may still exist even though websites cannot be found by name.

Literature & RFC Review

Tool	Main use	Limitation for PS-015
Speed-test tools	Speed and basic latency	Does not by itself explain the underlying condition.
Ping	Reachability, RTT, loss	Basic measurement; needs interpretation.
Traceroute	Shows network path	Does not provide the complete PS-015 diagnosis.
nslookup / dig	DNS testing	Focused mainly on DNS.
Wireshark	Packet capture and analysis	Powerful but technical; not a guided diagnosis workflow.
iPerf3	Throughput/load testing	Measurement tool rather than a complete diagnosis engine.

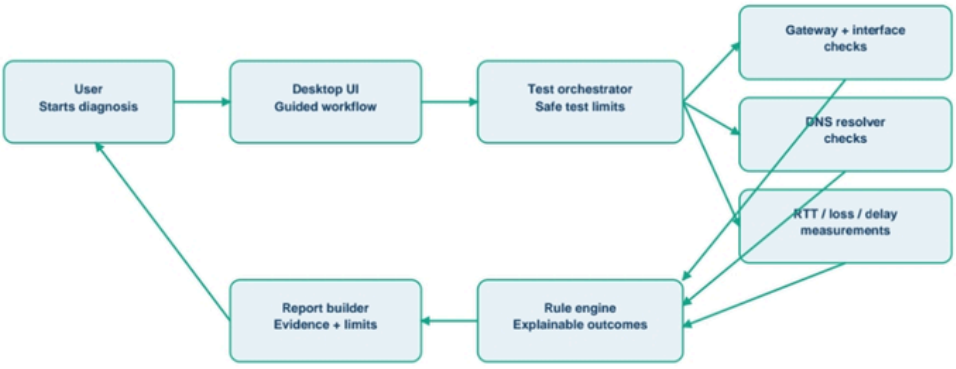
Proposed Solution

•Develop a desktop-based network diagnostic application that guides the user through a fixed set of tests. The application records measurements, evaluates them using transparent rules, produces a diagnosis only when supported by evidence, and generates a report.
•Interface and addressing check
•Gateway reachability test
•Beyond-gateway Internet reachability test
•DNS test using configured and alternate resolvers
•Packet-loss measurement
•Loaded latency / congestion measurement

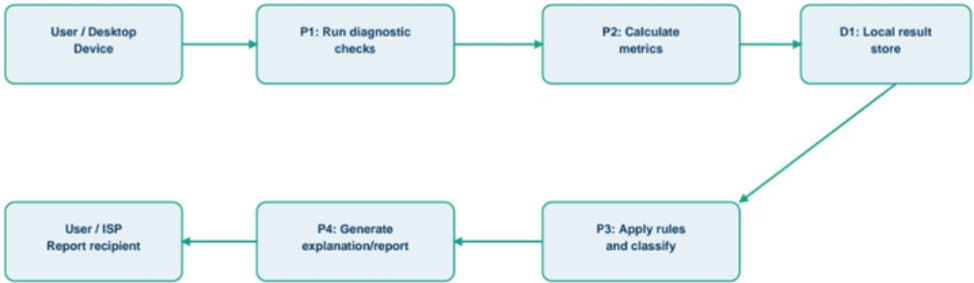
Optimization Proposed

Feature	Before	After
Testing	Basic speed and connectivity tests	Ping, DNS, and HTTP probes
Diagnosis	Manual interpretation of results	Rule-based fault identification
Detection	Limited fault localization	Checks for local network, DNS, packet loss, and congestion
Explainability	Results with limited reasoning	Evidence-backed findings with transparent rules
Uncertain cases	Potentially ambiguous conclusions	INCONCLUSIVE when evidence is insufficient
Reporting	Manual issue reporting	Escalation report with timestamps and measured metrics

Solution Architecture & Design



Data Flow Diagram



Deployment Timeline

Phase	Work
1	Learn basic networking: IP, gateway, DNS, ping, latency, packet loss.
2	Implement individual measurements.
3	Combine measurements into the guided workflow.
4	Implement rule-based decision engine.
5	Build desktop GUI and evidence/report generation.
6	Create controlled fault scenarios and test.
7	Prepare demo, diagrams, results and presentation.

Bibliography I

[1] V. Paxson, G. Almes, J. Mahdavi, and M. Mathis, “Framework for IP Performance Metrics,” RFC 2330, May 1998. https://www.rfc-editor.org/rfc/rfc2330

[2] A. Morton, “Guidelines for Creating New IP Performance Metrics,” RFC 7312, August 2014. Verify the exact official title and bibliographic details at https://www.rfc-editor.org/rfc/rfc7312 before submission.

Thank You for Your Attention!